Mock Exam 2026 Worked Walkthrough

Every sub-question solved in order · Method chosen · SPSS steps · Full answer

Q1 · Housing & Air Pollution (pollution.sav) — MLR · 7 pts

Q2 · US Cities & Homicide (output given) — MLR · 6 pts

Q3 · Chicken Demand (chicken.sav) — Time Series · 7 pts

For every sub-question this document shows: which section of the exam guide applies (green box), the reasoning and approach (orange box), the exact SPSS menu steps (blue box), and the complete worked answer with the real output numbers (green answer box).

QUESTION 1 — Housing & Air Pollution · pollution.sav · 7 points

Goal: model median housing value (CMEDV) using all available predictors. NOx must be logged. Region is a 4-category variable requiring dummy coding. $n = 506$ neighbourhoods.

Q1a Write down a theoretical equation of the model

Question asks: Write down a theoretical equation of the model.

Guide §1.1 + 1.2 + 1.3 — Reading the Question · Dummy Variables · Log Transformations

The key flags: (1) NOx must be logged → linear-log model. (2) Region has $k=4$ categories → create $k-1 = 3$ dummies, omitting one reference category. A theoretical equation lists all β coefficients, defines the dummies, and states the error term.

✎ Approach — Which method and why

- List every predictor. Flag Region as categorical (4 levels: NE, NW, South, Center).
- Choose Center as reference (any category works — state which you chose).
- Create 3 dummies: DNE (North East), DNW (North West), DS (South).
- Log-transform NOx: call it lnNOx. This creates a linear-log specification for that predictor.
- Write the equation with $\beta_0 \ldots \beta_{10}$, include ε, state the assumptions.

⌨ SPSS: Create Log Variable, Dummies, Run Regression

```
1. Transform → Compute Variable… Target: LN_NOx Expression: LN(NOx) OK.
2. Transform → Compute Variable… Target: DNE Expression: (Region = 1) OK.
3. Transform → Compute Variable… Target: DNW Expression: (Region = 2) OK.
4. Transform → Compute Variable… Target: DS Expression: (Region = 3) OK.
5. (Center = 4 is the reference — do NOT create a dummy for it.)
6. Analyze → Regression → Linear…
7. Dependent: CMEDV Independent(s): CRIM, LN_NOx, RM, AGE, DIS, RAD, PTRatio, DNE, DNW, DS
8. Click OK.
```

✓ Worked Answer

$CMEDV = \beta_0 + \beta_1 \cdot CRIM + \beta_2 \cdot lnNOx + \beta_3 \cdot RM + \beta_4 \cdot AGE + \beta_5 \cdot DIS + \beta_6 \cdot RAD$

$+ \beta_7 \cdot PTRatio + \beta_8 \cdot DNE + \beta_9 \cdot DNW + \beta_{10} \cdot DS + \varepsilon$

Assumptions: $\varepsilon \sim N(0, \sigma^2)$ independent; predictors not perfectly multicollinear;

ε uncorrelated with each predictor.

Dummy definitions (reference category = Center):

DNE = 1 if neighbourhood is in North East, 0 otherwise

DNW = 1 if neighbourhood is in North West, 0 otherwise

DS = 1 if neighbourhood is in South, 0 otherwise

Why 3 dummies for 4 categories? Rule: always k−1. Including all 4 would cause

perfect multicollinearity (DNE + DNW + DS + DCenter ≡ 1 for every observation).

Q1b Discuss the effect of pollution on housing values

Question asks: Discuss the effect of pollution on the housing value based on your fitted model.

Guide §1.3 + 1.5 — Log Transformation Interpretation · Testing Individual Significance

NOx is logged → linear-log model. Interpretation rule: 1% increase in X → β/100 units change in Y.
Also test significance: read Sig. column for lnNOx from the Coefficients table.

✎ Approach — Which method and why

- Locate the lnNOx row in the Coefficients table. Read B (= −16.041).
- Apply the linear-log rule: 1% increase in NOx → −16.041/100 = −0.160 thousand dollars.
- Convert to dollars: −0.160 × \$1000 = −\$160.41 decrease in median housing value.
- Check significance: t = −5.572, Sig. = 0.000 < 0.05 → effect is statistically significant.

Predictor	B	Std. Error	t	Sig.	Interpretation
lnNOx	−16.041	2.879	−5.572	0.000	Significant at 5%

✓ Worked Answer

Model type: Linear-Log (Y is in levels, X is logged).

Rule: 1% increase in X → β/100 units change in Y.

Interpretation: A 1% increase in NOx concentration decreases median housing value

by −16.041 / 100 = −0.160 thousand dollars = −\$160.41 on average, holding all other predictors constant.

Significance: $t = -5.572$, $p = 0.000 < 0.05$.

We reject H_0: $\beta\_lnNOx = 0$. The effect of pollution is statistically significant.

Direction makes economic sense: higher pollution → lower desirability → lower house prices.

Q1c Does Region have a significant effect on CMEDV? (one hypothesis test)

Question asks: Does the variable Region have a significant effect on CMEDV? Carry out one hypothesis test (H_0, H_1, test statistic, conclusion, assumptions).

Guide §1.6 — Joint F-test — Testing a Group of Variables Together

Region produces 3 dummies. Testing them individually with 3 separate t-tests inflates Type I error. The correct method is one joint F-test: H_0: $\beta\_DNE = \beta\_DNW = \beta\_DS = 0$. Use SPSS block method to get the R^2-change F-statistic automatically.

✎ Approach — Which method and why

- $J = 3$ restrictions (3 dummies), $k = 10$ (full model), $n = 506$.
- Unrestricted model: all 10 predictors (including 3 dummies). $R^2\_u = 0.652$.
- Restricted model: 7 predictors (dummies removed). $R^2\_r = 0.649$.
- $F = [(R^2\_u - R^2\_r)/J] / [(1 - R^2\_u)/(n - k - 1)]$.
- Use p-value from SPSS Change Statistics to make the decision.

⌨ SPSS: Joint F-test via Block Method (R-square Change)

```
1. Analyze → Regression → Linear…
2. Block 1: Dependent CMEDV, Independent(s): CRIM LN_NOx RM AGE DIS RAD PTRatio (no dummies).
3. Click Next → Block 2: add DNE, DNW, DS.
4. Click Statistics… → check R-square change → Continue.
5. Click OK. Read the Model Summary table: row 2, column Sig. F Change.
```

✓ Worked Answer

H_0: $\beta\_DNE = \beta\_DNW = \beta\_DS = 0$ (Region has no effect — restricted model)

H_1: At least one of β_DNE, β_DNW, $\beta\_DS \neq 0$ (Region has an effect — full model)

Assumption: errors are normally distributed (for exact F-distribution).

Test statistic (R^2-change F):

$F = [(0.652 - 0.649) / 3] / [(1 - 0.652) / (506 - 10 - 1)] = 0.001 / 0.000703 = 1.463$

From SPSS Change Statistics: F Change = 1.463, df1 = 3, df2 = 495, Sig. F Change = 0.224.

Decision: $p = 0.224 > 0.05$ → Fail to reject H_0.

Conclusion: The region dummies do not jointly improve the model significantly.

Region does NOT have a statistically significant effect on CMEDV at the 5% level.

Q1d Set up a residual plot and discuss

Question asks: Set up a residual plot and discuss this.

Guide §1.7 — Residual Plot Analysis

Plot standardized residuals (ZRESID) on Y-axis vs standardized predicted values (ZPRED) on X-axis. A good plot shows a random horizontal cloud. Look for: fan shape (heteroskedasticity), curved arc (non-linearity), isolated outliers.

⌨ SPSS: Generate Residual Plot

```
1. Analyze → Regression → Linear… (same model as Q1a, all predictors including dummies).
2. Click Plots… → set Y: ZRESID, X: ZPRED → Continue.
3. Click Save… → check Unstandardized under Residuals and Unstandardized under Predicted Values → Continue.
4. Click OK. The scatter plot appears in the Output Viewer.
5. Alternatively: Graph → Chart Builder → Scatter → drag PRE_1 to x-axis, RES_1 to y-axis → OK.
```

✓ Worked Answer

What to look for: a random horizontal cloud around zero → all assumptions hold.

What the actual plot shows (from official solution):

- Several neighbourhoods show outlying behaviour — isolated points far from zero.
- Some points form a 'line pattern', suggesting subgroups behave differently.
- For very low and very high predicted values, residuals tend to be positive

→ mean of errors is not zero for all predictions → Assumption A1 potentially violated.

• The spread may widen for larger fitted values (possible heteroskedasticity).

Conclusion: The residual plot is NOT ideal. Some regression assumptions appear violated.

The 506 neighbourhoods may not be homogeneous enough for a single linear model.

Key rule: describe WHAT you see and NAME the assumption violated.

Fan/cone → A2 violated (non-constant variance).

Curved arc → A1 violated (model misspecified).

Outliers → investigate individually (can distort estimates).

Q1e Write the auxiliary model for the White test

Question asks: Write down the auxiliary model you would use to carry out the White test (you do not have to carry out the test).

Guide §1.8 — White Test for Heteroskedasticity — Auxiliary Model

The auxiliary regression regresses $\hat{e}^2$ on: all original predictors, the square of each CONTINUOUS predictor, and all cross-products of pairs of predictors. Never square dummies ($D^2 = D$). Never include dummy×dummy cross-products.

✎ Approach — Which method and why

• Continuous variables: CRIM, lnNOx, RM, AGE, DIS, RAD, PTRatio (7 variables).

• Dummies: DNE, DNW, DS — do NOT square these.

• Add squared terms for each continuous variable: $CRIM^2$, $lnNOx^2$, RM^2, AGE^2, DIS^2, RAD^2, $PTRatio^2$.

• Add cross-products for every pair of continuous variables: 7 choose 2 = 21 pairs.

• Add cross-products of each continuous variable with each dummy: 7 × 3 = 21 pairs.

• Total regressors in auxiliary model (excluding constant) = 10 + 7 + 21 + 21 = 59.

• Note: df for the White test statistic = 59. $W \sim \chi^2(59)$.

✓ Worked Answer

Let $\hat{e}$ = residuals from the Q1a model. Auxiliary regression:

$$\hat{e}^2 = \beta_0 + \beta_1 CRIM + \beta_2 lnNOx + \beta_3 RM + \beta_4 AGE + \beta_5 DIS + \beta_6 RAD + \beta_7 PTRatio$$

$$+ \beta_8 DNE + \beta_9 DNW + \beta_{10} DS$$

$+ \alpha_1 CRIM^2 + \alpha_2 lnNOx^2 + \alpha_3 RM^2 + \alpha_4 AGE^2 + \alpha_5 DIS^2 + \alpha_6 RAD^2 + \alpha_7 PTRatio^2$

$+ \gamma_1 CRIM \cdot lnNOx + \gamma_2 CRIM \cdot RM + \gamma_3 CRIM \cdot AGE + \gamma_4 CRIM \cdot DIS$

$+ \gamma_5 CRIM \cdot RAD + \gamma_6 CRIM \cdot PTRatio + \gamma_7 CRIM \cdot DNE + \gamma_8 CRIM \cdot DNW + \gamma_9 CRIM \cdot DS$

$+ \gamma_{10} lnNOx \cdot RM + \gamma_{11} lnNOx \cdot AGE + \gamma_{12} lnNOx \cdot DIS + \gamma_{13} lnNOx \cdot RAD$

$+ \gamma_{14} lnNOx \cdot PTRatio + \gamma_{15} lnNOx \cdot DNE + \gamma_{16} lnNOx \cdot DNW + \gamma_{17} lnNOx \cdot DS$

+ [all remaining continuous pairs: RM·AGE, RM·DIS, RM·RAD … DIS·RAD … etc.]

+ [continuous × dummy pairs: RM·DNE, RM·DNW, RM·DS … PTRatio·DS] + ε

Key rules: (1) Never square a dummy — $D^2 = D$. (2) Never cross dummy × dummy.

(3) $W = n \times R^2\_aux \sim \chi^2(df)$ where df = number of regressors above (excluding constant).

Note: the exam note says 'the model will probably be shorter' — if fewer predictors

are in the original model, the White auxiliary is much more manageable.

Q1f Omitted variable Indus — what does this mean for the lnNOx coefficient?

Question asks: Harrison and Rubinfield indicate that Indus (proportion of non-retail business acres) should also be included. You did not include it. What does this tell you about the estimate for the coefficient of lnNOx?

Guide §1.9 — Omitted Variable Bias

sign(bias) = sign(β_Indus) × sign(r(lnNOx, Indus)). Need to determine: (1) what sign would β_Indus have in the full model? (2) are lnNOx and Indus positively or negatively correlated?

✎ Approach — Which method and why

- Step 1 — Sign of β_Indus: Indus measures proportion of industrial land. More industry → less desirable neighbourhood → lower house prices → β_Indus < 0 (negative).
- Step 2 — Sign of r(lnNOx, Indus): Industrial areas produce more NOx pollution. Therefore lnNOx and Indus are positively correlated → r > 0.
- Step 3 — Bias direction: sign(bias) = sign(β_Indus) × sign(r) = (−) × (+) = negative.
- Negative bias means our estimate for β_lnNOx is TOO LOW (more negative than the true value).

Omitted variable Z	sign(β_Z)	sign(r(lnNOx, Z))	sign(bias) = product	β_lnNOx estimate is…
Indus	Negative (−)	Positive (+)	(−)×(+) = Negative	Too low (underestimated)

✓ Worked Answer

Indus is an omitted variable in our model. Omitted variables bias the coefficients

of correlated included variables.
sign(β_Indus) = negative (industrial areas reduce house values).
sign(r(lnNOx, Indus)) = positive (industrial areas also have more NOx).
Bias on β_lnNOx = (−) × (+) = negative bias.
Conclusion: our estimated coefficient for lnNOx (−16.041) is UNDERESTIMATED.
The true effect of pollution is more negative than −16.041 — we are understating
how much pollution harms housing values.
(In other words: some of the housing value reduction we attributed to pollution
is actually caused by the industrial character of the area, not NOx alone.)

QUESTION 2 — US Cities & Homicide Rates · Output provided · 6 points

No dataset to open — all answers are based on the SPSS output tables provided in the question. n = 100 cities. Dependent variable: HOMOCIDE. Predictors: SUN, INHABITANTS_log, POLICE, COAST.

Q2a What do you conclude from the VIF scores?

Question asks: What do you conclude from the VIF scores in the table?

Guide §1.10 — VIF and Multicollinearity

VIF = $1/(1 - R^2\_j)$. Threshold: VIF > 5 = concerning, VIF > 10 = severe. Read the Collinearity Statistics columns in the SPSS Coefficients table.

Variable	Tolerance	VIF	Assessment
SUN	0.890	1.123	Well below 5 — no problem
INHABITANTS_log	0.919	1.088	Well below 5 — no problem
POLICE	0.904	1.106	Well below 5 — no problem

✓ Worked Answer

All VIF scores are well below the rule-of-thumb threshold of 5.

Highest VIF = 1.123 (SUN) — very close to 1, indicating almost no multicollinearity.

Conclusion: There is no indication of (quasi-)multicollinearity in this model.

The coefficient estimates are reliable and the standard errors are not inflated.

Q2b Auxiliary regression to determine Tolerance for SUN

Question asks: Set up the equation of a suitable auxiliary regression model that will allow you to determine the Tolerance score for SUN. Explain how you determine the tolerance.

Guide §1.10 — VIF Formula: Tolerance = $1 - R^2\_j$

Tolerance for variable j = $1 - R^2$ from regressing X_j on all other X's. So: regress SUN on INHABITANTS_log and POLICE; read R^2; Tolerance(SUN) = $1 - R^2$.

✓ Worked Answer

Auxiliary regression model:

$\text{SUN} = \alpha_0 + \alpha_1 \cdot \text{INHABITANTS\_log} + \alpha_2 \cdot \text{POLICE} + \varepsilon$

Run this regression in SPSS. Read R^2_aux from the Model Summary table.

Tolerance(SUN) = 1 − R^2_aux

VIF(SUN) = 1 / Tolerance(SUN) = 1 / (1 − R^2_aux)

Verification: from the table, Tolerance(SUN) = 0.890, so R^2_aux ≈ 0.110.

This means only 11% of the variation in SUN is explained by the other two predictors

→ SUN contributes mostly independent information → low multicollinearity.

Q2c Is R^2 of Model 2 significantly larger than R^2 of Model 1?

Question asks: Carry out a suitable hypothesis test to investigate whether the R^2 of Model 2 is significantly larger than the R^2 of Model 1 (no need to check assumptions).

Guide §1.6 — Joint F-test — Comparing Two Nested Models

Model 2 adds POLICE to Model 1. So: full model = Model 2, restricted = Model 1, J = 1. This is a joint F-test (with J=1 it equals the individual t-test squared, but framing as F is cleaner).

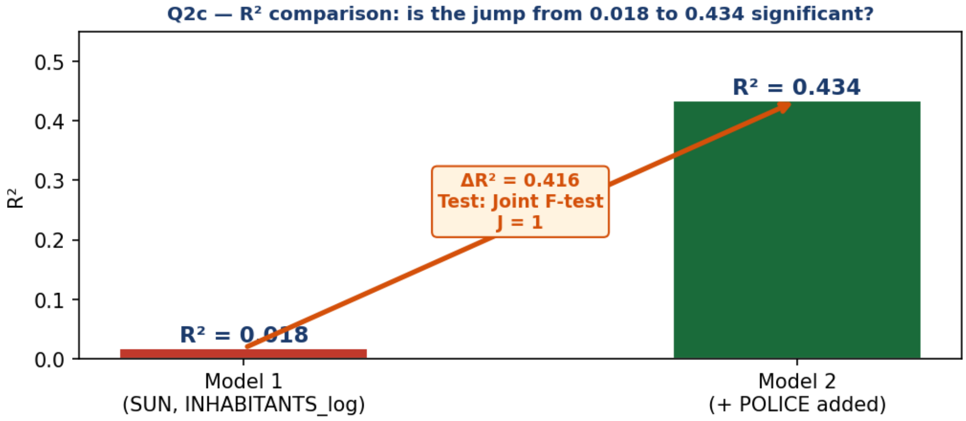

Figure Q2c — R^2 jumps from 0.018 to 0.434 when POLICE is added. The joint F-test tells us whether this increase is statistically significant.

✎ Approach — Which method and why

- Full (unrestricted) model = Model 2: HOMOCIDE = $\beta_0 + \beta_1$SUN + β_2INHABITANTS_log + β_3POLICE → R^2_u = 0.434.
- Restricted model = Model 1: HOMOCIDE = $\beta_0 + \beta_1$SUN + β_2INHABITANTS_log → R^2_r = 0.018.
- J = 1 (one restriction: β_POLICE = 0). n = 100. k = 3 (predictors in full model).
- F = [(R^2_u − R^2_r)/J] / [(1 − R^2_u)/(n − k − 1)].

✓ Worked Answer

H_0: β_POLICE = 0 (adding POLICE does not improve the model — Model 1 is sufficient)

H_1: β_POLICE ≠ 0 (POLICE improves the model — Model 2 is better)

Calculation:

```
F = [(0.434 − 0.018) / 1] / [(1 − 0.434) / (100 − 3 − 1)]
= 0.416 / (0.566 / 96)
= 0.416 / 0.005896
= 70.56
```

p-value = P(F ≥ 70.56) = 4.01×10^{-13} ≈ 0.000 (essentially zero).

Decision: p << 0.05 → Reject H_0.

Conclusion: The R^2 of Model 2 is significantly larger. Adding POLICE significantly

improves the model's explanatory power.

Q2d Why is the INHABITANTS_log coefficient different in both models?

Question asks: Explain how it is possible that the estimated effect of the number of inhabitants is different in both models, although both models are fitted using the same sample.

Guide §1.9 — Omitted Variable Bias — Ceteris Paribus Interpretation

When a relevant variable (POLICE) is left out of Model 1, its effect gets absorbed into the remaining coefficients. The coefficient on INHABITANTS_log in Model 1 is biased because POLICE is correlated with INHABITANTS_log.

Model	INHABITANTS_log coefficient	POLICE included?
Model 1 (restricted)	B = +0.105	No — POLICE omitted
Model 2 (full)	B = −0.267	Yes — POLICE included

✓ Worked Answer

In Model 2, the coefficient of INHABITANTS_log (−0.267) measures the effect of city size

on homicide rate HOLDING THE NUMBER OF POLICE OFFICERS CONSTANT (ceteris paribus).

In Model 1, POLICE is not in the model, so its effect is not held constant.

Any variation in homicide that is actually caused by POLICE gets incorrectly

attributed to INHABITANTS_log instead.

This is omitted variable bias: omitting POLICE biases the INHABITANTS_log coefficient upward (from −0.267 to +0.105, a swing of +0.372) because POLICE is correlated with INHABITANTS_log and has a positive effect on homicide.

Q2e Positive or negative correlation between INHABITANTS_log and POLICE?

Question asks: Do you expect a positive or negative correlation between INHABITANTS_log and POLICE? Explain using the output (not based on common sense).

Guide §1.9 — Omitted Variable Bias — Reverse-Engineering the Correlation Sign

We know the bias direction from the output (compare coefficients across models). Bias = sign(β_POLICE) × sign(r(INHABITANTS_log, POLICE)). Solve for sign(r) given the other two known signs.

✎ Approach — Which method and why

- Step 1: From Model 2, β_POLICE = +0.065 > 0 → POLICE has a positive effect on HOMOCIDE.
- Step 2: The bias on INHABITANTS_log = coefficient in Model 1 MINUS coefficient in Model 2 = +0.105 − (−0.267) = +0.372 → POSITIVE bias.
- Step 3: sign(bias) = sign(β_POLICE) × sign(r(INHABITANTS_log, POLICE)).
- (+0.372) = (+0.065) × sign(r) → sign(r) = positive.

✓ Worked Answer

From Model 2: POLICE has a significant positive effect on HOMOCIDE (B = 0.065, $p < 0.001$).

Comparing the two models, the INHABITANTS_log coefficient changed from +0.105 (Model 1) to −0.267 (Model 2) — a positive bias of +0.372 was present in Model 1.

Using the bias formula: sign(bias) = sign(β_POLICE) × sign(r(INHABITANTS_log, POLICE))

positive = (+) × sign(r) → sign(r) must be POSITIVE.

Conclusion: There is a POSITIVE correlation between INHABITANTS_log and POLICE.

(Larger cities tend to have more police officers — consistent with common sense too.)

Q2f Discuss the effect of SUN on number of inhabitants — with interaction

Question asks: Discuss the effect of the variable SUN on the number of inhabitants in a city. Model: INHABITANTS_log = $\beta_0 + \beta_1$SUN + β_2COAST + β_3(SUN×COAST) + ε. Coefficients: Constant 7.779, SUN 0.000 (p=1.000), COAST 0.150 (p=0.000), SUN×COAST 0.020 (p=0.000).

Guide §1.3 + 1.4 — Log-Linear Interpretation + Interaction Terms

The dependent variable is INHABITANTS_log = ln(INHABITANTS), making this a log-linear model. Rule for log-linear: 1-unit increase in X → β × 100% change in Y. The interaction term SUN×COAST means the effect of SUN DEPENDS ON whether the city is coastal. Split into two equations: one for COAST=0, one for COAST=1.

✎ Approach — Which method and why

- The model is: ln(INHABITANTS) = 7.779 + 0.000·SUN + 0.150·COAST + 0.020·(SUN×COAST).
- Split by COAST value to isolate the SUN effect in each group:
- Non-coast (COAST=0): ln(INHABITANTS) = 7.779 + 0.000·SUN → SUN slope = 0.000.
- Coast (COAST=1): ln(INHABITANTS) = (7.779+0.150) + (0.000+0.020)·SUN = 7.929 + 0.020·SUN.
- Since Y = ln(INHABITANTS): interpret using log-linear rule → β × 100% change per 1-unit X.
- Check significance: SUN alone p=1.000; SUN×COAST p=0.000.

City type	Equation	SUN effect	Significant?
Non-coast (COAST=0)	ln(INHABITANTS) = 7.779 + 0.000·SUN	0.000×100% = 0%	No (p=1.000)
Coast (COAST=1)	ln(INHABITANTS) = 7.929 + 0.020·SUN	0.020×100% = +2% per day	Yes (p=0.000)

✓ Worked Answer

The effect of SUN depends on whether the city is coastal (interaction term).

Non-coast cities: SUN has NO significant effect on the number of inhabitants.

The coefficient on SUN is 0.000 (p = 1.000) — cannot reject H_0 = no effect.

Coast cities: SUN has a significant POSITIVE effect.

Each extra day of sunshine → 0.020 × 100% = 2% increase in the number of inhabitants.

(Because Y = ln(INHABITANTS), the log-linear interpretation applies: β × 100%.)

The intercept shift for coast cities: COAST = 0.150 → coast cities have on average

e^0.150 ≈ 16% more inhabitants than non-coast cities at the same level of SUN.

QUESTION 3 — Chicken Demand · chicken.sav · 7 points

Goal: find a suitable time-series model for per-capita chicken consumption (q) as a function of real price (p). Always follow the Three-Step Framework: Stationarity → Fit → LMSC check.

Q3a Check whether p and q are stationary — Dickey-Fuller test

Question asks: Check whether p and q are stationary by carrying out suitable hypothesis tests. (H_0/H_1, test statistic, conclusion.)

Guide §2.2 — Step 1 — Dickey-Fuller Test for Stationarity

First plot the series. Both q and p show upward trends → use DF Version 2 (with trend variable). Critical value = −3.41 at 5%. NEVER use the Sig. p-value from SPSS for this test — compare the t-statistic for the LAG variable directly to −3.41.

✎ Approach — Which method and why

- Time plots of both q and p show a clear upward trend → select DF Version 2.
- Create DIFF_q = q − LAG(q,1) and LAG_q = LAG(q,1).
- Run: Dependent = DIFF_q, Independent = LAG_q AND Year (trend variable).
- Read t-statistic for LAG_q. Compare to −3.41. Ignore the Sig. column.
- Repeat exactly for p: DIFF_p, LAG_p, regression with LAG_p and Year.

⌨ SPSS: DF Test Version 2 (with trend) for Levels

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1. Transform → Compute: DIFF_q = q − LAG(q,1) | LAG_q = LAG(q,1).
2. Transform → Compute: DIFF_p = p − LAG(p,1) | LAG_p = LAG(p,1).
3. Analyze → Regression → Linear: Dependent=DIFF_q, Independent=LAG_q, Year. OK.
4. Read t for LAG_q. Compare to −3.41 (NOT the Sig. column!).
5. Analyze → Regression → Linear: Dependent=DIFF_p, Independent=LAG_p, Year. OK.
6. Read t for LAG_p. Compare to −3.41.
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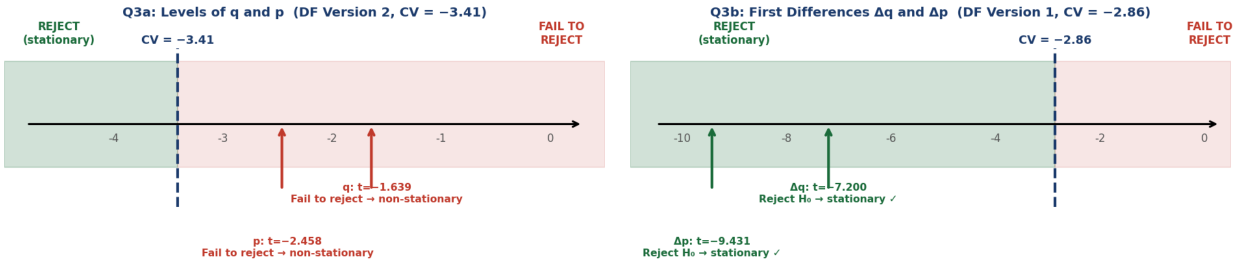

Figure Q3a/b — Left: DF results for levels q and p (Version 2, CV=−3.41). Both t-statistics are above the critical value → non-stationary. Right: DF results for first differences (Version 1, CV=−2.86). Both t-statistics are far below the critical value → stationary.

✓ Worked Answer

Both series show a trend in the time plot → DF Version 2 (with trend).

Critical value = −3.41 (5% level, Version 2).

Is q stationary?

H_0: $\alpha_1 = 0$ (unit root — non-stationary) H_1: $\alpha_1 < 0$ (stationary)

Auxiliary model: $\Delta q_t = \alpha_0 + \alpha_1 q_{t-1} + \alpha_2 t + \varepsilon_t$

t-statistic for LAG_q = −1.639. −1.639 > −3.41 → NOT more negative than CV.

Decision: Fail to reject H_0. q is NOT stationary.

Is p stationary?

H_0: $\alpha_1 = 0$ H_1: $\alpha_1 < 0$

Auxiliary model: $\Delta p_t = \alpha_0 + \alpha_1 p_{t-1} + \alpha_2 t + \varepsilon_t$

t-statistic for LAG_p = −2.458. −2.458 > −3.41 → NOT more negative than CV.

Decision: Fail to reject H_0. p is NOT stationary.

Q3b Do the same for the first-order differences of p and q

Question asks: Do the same for the first order differences of p and q.

Guide §2.2 — Step 1 continued — DF Test on First Differences — Determining I(1)

Plot Δq and Δp. If they show no trend → use DF Version 1 (no trend). Critical value = −2.86. If both differences are stationary, both original series are I(1).

✎ Approach — Which method and why

- Time plots of Δq and Δp do NOT show a clear upward trend → use DF Version 1.
- Create DIFF2_q = DIFF_q − LAG(DIFF_q,1) and LAG_DIFF_q = LAG(DIFF_q,1).
- Run: Dependent = DIFF2_q, Independent = LAG_DIFF_q only (no Year this time).
- Read t for LAG_DIFF_q. Compare to −2.86. Repeat for p.

⌨ SPSS: DF Test Version 1 (no trend) for First Differences

```
1. Transform → Compute: DIFF2_q = DIFF_q − LAG(DIFF_q,1) | LAG_DIFF_q = LAG(DIFF_q,1).
2. Transform → Compute: DIFF2_p = DIFF_p − LAG(DIFF_p,1) | LAG_DIFF_p = LAG(DIFF_p,1).
```

3. Analyze → Regression → Linear: Dependent=DIFF2_q, Independent=LAG_DIFF_q only. OK.

4. Read t for LAG_DIFF_q. Compare to −2.86.

5. Analyze → Regression → Linear: Dependent=DIFF2_p, Independent=LAG_DIFF_p only. OK.

6. Read t for LAG_DIFF_p. Compare to −2.86.

✓ Worked Answer

First differences show no obvious trend → DF Version 1 (no trend).

Critical value = −2.86 (5% level, Version 1).

Is Δq stationary?

$H_0: \alpha_1 = 0$ $H_1: \alpha_1 < 0$

Auxiliary model: $\Delta\Delta q_t = \alpha_0 + \alpha_1 \Delta q_{t-1} + \varepsilon_t$

t-statistic for LAG_DIFF_q = −7.200. −7.200 < −2.86 → more negative than CV.

Decision: Reject H_0. Δq IS stationary ✓

Is Δp stationary?

$H_0: \alpha_1 = 0$ $H_1: \alpha_1 < 0$

Auxiliary model: $\Delta\Delta p_t = \alpha_0 + \alpha_1 \Delta p_{t-1} + \varepsilon_t$

t-statistic for LAG_DIFF_p = −9.431. −9.431 < −2.86 → more negative than CV.

Decision: Reject H_0. Δp IS stationary ✓

Conclusion: Both q and p are I(1) — they have a unit root in levels but are stationary

after first differencing. This drives the model choice in Q3c.

Q3c Levels or first differences for the static model?

Question asks: To set up a static model for demand as a function of price, do you prefer a model based on the original variables p and q, or on the first order differences? Explain.

Guide §2.3 — Choosing the Right Variables — Stationarity Rule

A fundamental rule: never regress a non-stationary series on another non-stationary series without checking for cointegration. Using non-stationary series in levels risks spurious regression (high R^2, but relationship is meaningless). When both series are I(1) and not cointegrated, use first differences.

✓ Worked Answer

From Q3a: both q and p are NON-STATIONARY in levels.

From Q3b: both Δq and Δp ARE stationary (I(1) series).

Preferred choice: use FIRST DIFFERENCES (Δq and Δp).

Reason: regressing non-stationary q on non-stationary p risks SPURIOUS REGRESSION.

A spurious regression produces a high R^2 and significant t-statistics even when

the two series have no true relationship — the apparent relationship is driven by

the common upward drift in both series, not by a genuine economic link.

By using first differences (which are stationary), we eliminate the unit root

problem and can interpret the results as a genuine causal relationship.

Q3d Fit the static model and test price significance

Question asks: Set up the static model you preferred in part c and carry out a suitable test to check whether the price difference has a significant effect on demand difference.

Guide §2.3 + 1.5 — Static Model · Individual t-test

Static model: $\Delta q_t = \beta_0 + \beta_1 \Delta p_t + \varepsilon_t$. Test H_0: $\beta_1 = 0$ using the t-test p-value (Sig. column) from the SPSS Coefficients table.

⌨ SPSS: Run the Static First-Difference Model

```
1. Analyze → Regression → Linear…
2. Dependent: DIFF_q | Independent(s): DIFF_p
3. Click OK. Read the Coefficients table row for DIFF_p: B, t, Sig.
```

	B	Std. Error	t	Sig.
(Constant)	0.678	0.129	5.264	0.000
DIFF_p (= Δp)	−2.596	1.057	−2.456	0.018

✓ Worked Answer

Model specification: $\Delta q_t = \beta_0 + \beta_1 \cdot \Delta p_t + \varepsilon_t$

Fitted model: $\Delta q = 0.678 - 2.596 \cdot \Delta p$

Hypothesis test for price effect:

H_0: $\beta_1 = 0$ (price change has no effect on demand change)

H_1: $\beta_1 \neq 0$ (price change has a significant effect)

Test statistic: $t = -2.456$

p-value = 0.018 < 0.05 → Reject H_0.

Conclusion: The change in price has a statistically significant effect on the change in demand at the 5% level.

Interpretation: A 1-unit increase in the price difference leads to a 2.596-unit decrease in the demand difference — as expected (higher price → lower consumption).

Q3e Test for autocorrelation — LMSC test on the static model

Question asks: Consider the model from Q3d. Carry out a hypothesis test to check for autocorrelation. (H_0/H_1, auxiliary model, test statistic, p-value, conclusion.)

Guide §2.6 — Step 3 — LMSC Autocorrelation Test

Save residuals from the Q3d model. Create LAG_RES. Run auxiliary regression: RES_1 = α_0 + α_1·DIFF_p + α_2·LAG_RES + u. LM = n × R^2_aux ~ $\chi^2(1)$. Critical value = 3.841. NEVER use Durbin-Watson here.

⌨ SPSS: LMSC Test — Auxiliary Regression on Residuals

```
1. In the Q3d regression dialog: click Save… → check Unstandardized Residuals → Continue → OK.
2. SPSS creates variable RES_1 in the dataset.
3. Transform → Compute: LAG_RES1 = LAG(RES_1, 1) OK.
4. Analyze → Regression → Linear…
5. Dependent: RES_1 | Independent(s): DIFF_p AND LAG_RES1
6. Click OK. Record R²_aux from Model Summary.
7. n_aux = Total df + 1 from the ANOVA table.
8. Compute: LM = n_aux × R²_aux. Compare to 3.841.
```

Auxiliary model R^2	n_aux (Total df + 1)	LM = n × R^2	Critical value	Decision
R^2_aux = 0.011943	50 × 1 + 0 = 50	50 × 0.011943 = 0.597	3.841	0.597 < 3.841 → Fail to reject

✓ Worked Answer

H_0: $\rho = 0$ (no autocorrelation — Assumption A3 satisfied)

H_1: $\rho \neq 0$ (autocorrelation present — A3 violated)

Auxiliary regression: $e_t = \alpha_0 + \alpha_1 \cdot \Delta p_t + \alpha_2 \cdot e_{t-1} + u_t$

R^2_aux = 0.011943. n_aux = 50 (from ANOVA Total df + 1).

LM = 50 × 0.011943 = 0.597

p-value = $P(\chi^2(1) \geq 0.597) = 0.4397 > 0.05$.

Decision: Fail to reject H_0.

Conclusion: No significant evidence of autocorrelation. Assumption A3 is

plausibly satisfied. The static model residuals are not serially correlated.

Q3f Distributed lag model — top-down selection from DL(3)

Question asks: Estimate a distributed lag model with maximum lag = 3. How many lags will you use in the final model? Explain with intermediate results.

Guide §2.4 — DL(q) — Top-Down Lag Selection

Start at q_max=3. At each step look at the Sig. of the HIGHEST lag in the model. If Sig. > 0.05 → drop that lag → refit at q−1. Stop when the highest remaining lag is significant (Sig. ≤ 0.05). NEVER drop an intermediate lag — structure must remain contiguous.

⌨ SPSS: Top-Down DL Lag Selection

```
1. Transform → Compute: LAG1_DIFF_p = LAG(DIFF_p, 1) OK.
2. Transform → Compute: LAG2_DIFF_p = LAG(DIFF_p, 2) OK.
3. Transform → Compute: LAG3_DIFF_p = LAG(DIFF_p, 3) OK.
4. Step 1 — Fit DL(3): Dependent=DIFF_q, Independent=DIFF_p, LAG1, LAG2, LAG3. Read Sig. of LAG3.
5. Step 2 — If LAG3 not significant: remove it. Fit DL(2): Dependent=DIFF_q, IVs=DIFF_p, LAG1, LAG2.
6. Step 3 — If LAG2 not significant: remove it. Fit DL(1): Dependent=DIFF_q, IVs=DIFF_p, LAG1.
7. Step 4 — If LAG1 not significant: remove it. Final model = static (DIFF_p only).
8. Stop as soon as the highest lag IS significant.
```

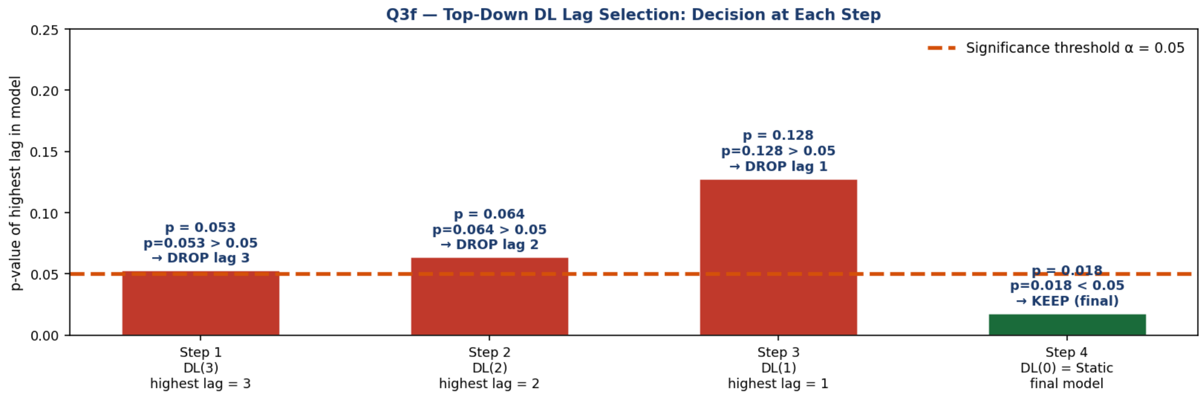

Figure Q3f — p-value of the highest lag at each step of top-down selection. Red bars = not significant ($p > 0.05$) → lag dropped. Green bar = final static model ($p = 0.018$ for DIFF_p). All three lags are dropped → optimal $q = 0$.

Step	Model	Highest lag	Sig. of highest lag	Decision
1	DL(3): DIFF_p + LAG1 + LAG2 + LAG3	LAG3	0.053 > 0.05	Drop LAG3
2	DL(2): DIFF_p + LAG1 + LAG2	LAG2	0.064 > 0.05	Drop LAG2
3	DL(1): DIFF_p + LAG1	LAG1	0.128 > 0.05	Drop LAG1
4	DL(0) = Static: DIFF_p only	DIFF_p	0.018 < 0.05	STOP — keep

✓ Worked Answer

Step 1 — DL(3): $\Delta q = 0.776 + (-2.214)\Delta p + 0.640\Delta p_{t-1} + (-1.552)\Delta p_{t-2} + 2.134\Delta p_{t-3}$

Sig. of LAG3 = 0.053 > 0.05 → NOT significant → drop LAG3.

Step 2 — DL(2): $\Delta q = 0.685 + (-1.750)\Delta p + 1.153\Delta p_{t-1} + (-2.089)\Delta p_{t-2}$

Sig. of LAG2 = 0.064 > 0.05 → NOT significant → drop LAG2.

Step 3 — DL(1): $\Delta q = 0.765 + (-2.090)\Delta p + 1.711\Delta p_{t-1}$

Sig. of LAG1 = 0.128 > 0.05 → NOT significant → drop LAG1.

Step 4 — DL(0) = Static: $\Delta q = 0.678 - 2.596\Delta p$

Sig. of DIFF_p = 0.018 < 0.05 → SIGNIFICANT → STOP.

Final answer: 0 lags are used in the final model.

The optimal model is the static model already fitted in Q3d.

The price effect is immediate only — no lagged price effects are significant.

Total multiplier = $\beta_0 = -2.596$ (same as immediate effect since $q = 0$).

★ General Exam Strategy — Key Reminders
• Q1-type (MLR): always check for categorical variables → need k−1 dummies. If a variable is logged, use the β/100 interpretation for linear-log.
• Q1c-type (joint test): the phrase 'does Region have a significant effect' means ONE joint F-test, not three separate t-tests. Use SPSS block method to get the F change automatically.
• Q1f-type (omitted variable): apply sign(β_Z) × sign(r(X,Z)) to find bias direction. Two negatives = positive bias = overestimate.
• Q2c-type (model comparison): the restricted model has FEWER variables; its SSE is always LARGER.
• Q2e-type (reverse bias): use the known bias direction and β_omitted sign to solve for r sign.
• Q3-type (time series): ALWAYS check stationarity first. Non-stationary series → use differences. NEVER use p-value from SPSS for DF test — compare t directly to −2.86 or −3.41.
• Q3e-type (LMSC): auxiliary regression must include ALL original X variables + lagged residual. $LM = n \times R^2\_aux \sim \chi^2(1)$. Critical value = 3.841.
• Q3f-type (DL top-down): only drop the HIGHEST lag. If lag 2 is significant keep lag 1 even if lag 1 is not — never skip a lag in the middle.